Image DNA

Intelligent Image Authenticity & Forgery Detection System

Project Documentation

Part 1: Project Planning & Management
Part 2: Literature Review
Part 3: Requirements Gathering

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Part 1: Project Planning & Management

1.1 Project Overview

Digital images are now trivial to forge. Graduation certificates, official documents, ID card photos and personal portraits can be edited in Photoshop (changing a number or a name) or regenerated with AI (face swapping, inpainting, fully synthetic faces). The human eye can no longer reliably tell an authentic image from a forged one, and there is no everyday tool that lets an ordinary person or an organisation check an image quickly and credibly.

Image DNA is a layered verification platform that works like an inspection checkpoint for images. Every image submitted passes through **four security barriers** in sequence, and the evidence from all of them is combined into one explainable verdict.

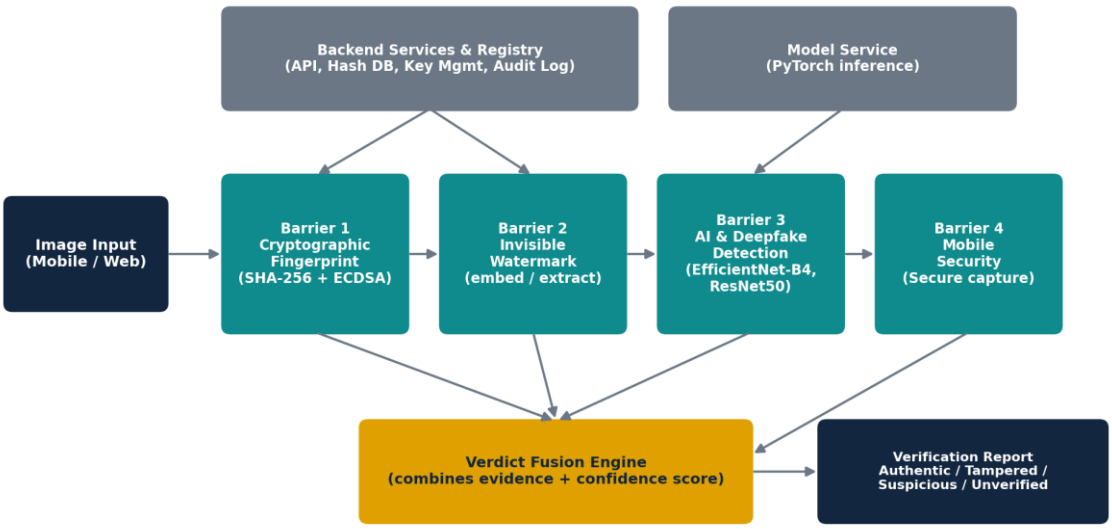


Figure 1: High-level architecture of Image DNA (four barriers + verdict fusion).

1.2 Problem Statement

- Image editing and generative-AI tools are widely available, cheap and increasingly realistic.
- Visual inspection and simple metadata (EXIF) checks are unreliable; metadata is trivially stripped or edited.
- Single-technique tools fail: a hash cannot flag an image it has never seen, a watermark can be removed, and an AI detector can be fooled or misjudge new generators.
- Organisations (universities, HR, banks, notaries) lack an accessible way to verify the authenticity of images submitted to them.

1.3 Vision, Goals and Objectives

Vision: make image authenticity checkable by anyone in seconds, with evidence that is cryptographic where possible and statistical where not, and always honest about its confidence.

ID	Objective (SMART)	Measure of completion
O1	Implement Barrier 1: SHA-256 fingerprinting with ECDSA digital signatures and a registry, detecting any single-bit change of a registered image.	100% detection of byte-level modification on the test set; signature verification in < 1 s.

ID	Objective (SMART)	Measure of completion
O2	Implement Barrier 2: invisible watermark embedding and extraction that survives common benign operations (JPEG recompression, resize, mild noise).	Watermark bit-accuracy >= 95% after JPEG Q70 and 50% resize on the test set; PSNR >= 38 dB (invisible).
O3	Train and evaluate Barrier 3: a deep-learning model that detects manipulated images, face swaps and AI-generated images *without* having seen the original.	AUC >= 0.95 on in-distribution test data; documented performance on unseen generators / datasets.
O4	Deliver Barrier 4: a hardened mobile app with secure capture/signing and app-integrity checks.	App passes the security checklist (OWASP MASVS-based) with no critical findings.
O5	Build a fusion engine that outputs a clear verdict (Authentic / Tampered / Suspicious / Unverified) with a confidence score and explanation.	Verdict returned in < 10 s per image on the reference server; report explains each barrier's contribution.

1.4 Project Scope

In scope

- Still images (JPEG/PNG) of faces, ID cards, certificates and general documents.
- Mobile app (verify + secure capture), REST backend, image registry, model-serving service.
- Detection of: byte-level tampering (hash/signature), watermark loss/damage, splicing/copy-move/retouching, face swap, and AI-generated faces/images.
- Verification report (on screen and as PDF) with per-barrier evidence.

Out of scope (version 1)

- Video and audio deepfake detection (listed as future work).
- Legal certification or admissibility of results in court.
- Detection guarantees against adaptive adversaries who know the model internals.
- Verification of the real-world truth of what a photo depicts (we verify the image, not the event).

1.5 Solution Overview: The Four Barriers

Barrier	Technique	What it answers	Key limitation (and how we cover it)
1. Fingerprint & Signature	SHA-256 hash, ECDSA (P-256) signature, optional perceptual hash (pHash), registry lookup	Is this exact file the one that was registered/signed by a trusted party?	Only works for registered images; any re-save changes the hash. Covered by perceptual hash + barriers 2 and 3.
2. Invisible Watermark	Frequency-domain (DWT-DCT-SVD) embedding; optional neural encoder/decoder	Was this image issued by our system, and is the mark intact?	Can be damaged by heavy edits/screenshots. Covered by barrier 1 and 3.
3. AI & Deepfake Detection	EfficientNet-B4 (transfer learning) with MTCNN/RetinaFace face extraction; ELA + ResNet50 for pixel-level tampering	Do internal image traces show manipulation or synthetic generation, even for never-seen images?	Probabilistic, may generalise poorly to new generators. Covered by confidence reporting and retraining plan.
4. Mobile Protection	Secure capture, Keystore/Secure Enclave signing, app attestation, TLS pinning, root/jailbreak detection	Was the image captured/submitted by a genuine, untampered app on a genuine device?	Cannot protect against a fully compromised device. Covered by server-side checks.

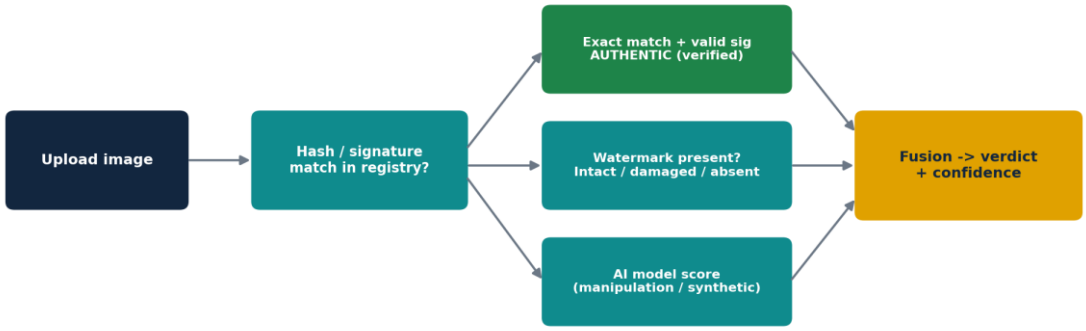


Figure 2: Verification flow and verdict fusion.

> **Design principle (honesty by design):** a registered image with a valid signature is verified with cryptographic certainty. An unregistered image can never be declared "100% authentic"; the system reports the strength of the evidence instead (Unverified or Suspicious with a confidence score). This avoids false assurance, which is the worst failure mode for a verification product.

1.6 Development Methodology

The team follows an **Agile / Scrum-style** process with two-week sprints (8 sprints over 16 weeks). Each sprint ends with a working increment and a short demo to the supervisor. Work is tracked with GitHub Issues and a GitHub Projects board (To do / In progress / Review / Done). Rules: feature branches, pull-request review by at least one other member, and CI running unit tests on every merge to main.

1.7 Work Plan and Timeline (16 Weeks)

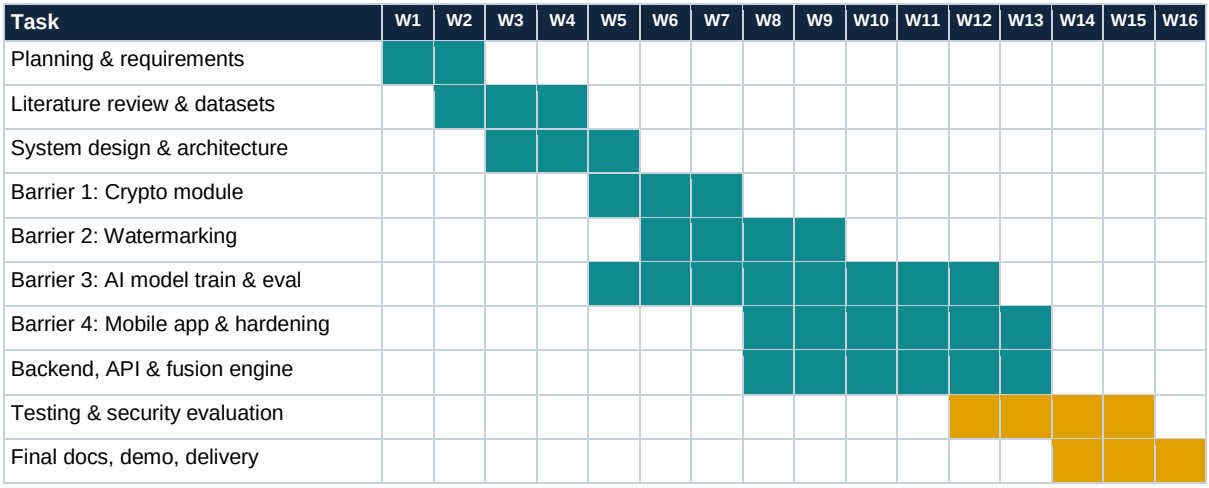


Figure 3: Project Gantt chart (weeks W1 to W16).

Phase	Weeks	Main deliverables
Planning & requirements	W1-2	Project plan, requirements document (this document, Parts 1 and 3)
Literature review & datasets	W2-4	Literature review (Part 2), datasets collected and cleaned, baseline experiments
System design	W3-5	Architecture, database schema, API contract, UI wireframes
Barrier 1: Crypto	W5-7	Hashing/signing library, key management, registry API, unit tests
Barrier 2: Watermark	W6-9	Embed/extract module, robustness test suite
Barrier 3: AI model	W5-12	Trained models, evaluation report (AUC, F1, cross-dataset), inference service

Phase	Weeks	Main deliverables
Barrier 4 & mobile app	W8-13	Mobile app, secure capture, attestation, hardening checklist
Backend & fusion	W8-13	REST API, fusion engine, report generator, Docker deployment
Testing & security evaluation	W12-15	Test report, performance benchmarks, security review
Final documentation & demo	W14-16	Final documentation, demo video, presentation, release v1.0

Milestones

Milestone	Week	Exit criterion
M1: Requirements & design approved	W5	Supervisor sign-off on documentation Parts 1-3 and architecture
M2: Barriers 1 and 2 working	W9	End-to-end sign + watermark + verify on test images
M3: AI model v1	W12	Model meets O3 on in-distribution test set
M4: Integrated beta	W13	Mobile app calls backend and shows fused verdict
M5: Final release	W16	All acceptance tests pass; documentation delivered

1.8 Team Structure and Responsibilities

Roles are proposed for the five team members and can be adjusted by the team. Every member also participates in testing and documentation.

Role	Member	Responsibilities
Project Manager / Business Analyst	Abdullah Mahmoud Fathy Fadl	Planning, sprint management, requirements, stakeholder communication, risk register, final documentation
Security & Backend Engineer	Ahmed Ali Fareeq El-Qazzaz	Barrier 1 (hashing, signatures, key management), registry, REST API, database, deployment
Image Processing Engineer	Mazen Saoudy Abdel Halim Morsy	Barrier 2 (watermarking), image pre-processing, robustness testing, perceptual hashing
AI / ML Engineer	Shahd Mohamed Mostafa El-Howeihy	Barrier 3: datasets, training, evaluation, model serving, explainability (heat-maps)
Mobile & QA Engineer	Shireen Mohamed Abdel Sattar Mostafa	Barrier 4 (mobile app, secure capture, attestation), UI/UX, test automation

RACI summary: the Project Manager is accountable for schedule and scope; each barrier owner is responsible for the design, implementation and tests of their barrier; the supervisor is consulted at every milestone.

1.9 Risk Management

#	Risk	Prob.	Impact	Mitigation / contingency
R1	AI model generalises poorly to unseen generators or datasets	High	High	Train on multiple datasets/generators; evaluate cross-dataset; report confidence, not binary truth; schedule retraining; fall back on barriers 1-2
R2	Insufficient or low-quality training data, class imbalance	Med	High	Use public benchmark datasets; augmentation; generate additional fakes with open tools; balanced sampling

#	Risk	Prob.	Impact	Mitigation / contingency
R3	Limited GPU/compute for training	Med	Med	Use transfer learning with pre-trained backbones; free cloud GPU tiers; transfer learning with EfficientNet-B4 and ResNet50; reduce image size / batch size or use mixed precision if memory is limited
R4	Watermark destroyed by screenshots, heavy compression or cropping	High	Med	Define supported operations explicitly; combine with perceptual hash; clear status "watermark damaged" instead of false alarm
R5	False positives wrongly accuse a genuine image	Med	High	Tune thresholds for low false-positive rate; three-level verdict with "Suspicious" band; human review guidance
R6	Private-key compromise or poor key management	Low	Critical	Keys in hardware-backed store or KMS; key rotation; revocation list; no keys in repository
R7	Privacy / legal concerns when handling IDs and faces	Med	High	Data minimisation; encryption at rest and in transit; retention limits; explicit consent; no training on user uploads without consent
R8	Team member unavailable / unequal workload	Med	Med	Pair ownership of each barrier; weekly sync; documented code; flexible task reallocation
R9	Scope creep (video, extra document types)	Med	Med	Scope frozen at M1; extras tracked as future work
R10	Integration problems between modules	Med	Med	Early API contracts; integration tests from W8; Docker Compose environment
R11	Adversarial attacks on the detector	Low	Med	Document as limitation; test basic attacks (resize, blur, JPEG); plan adversarial training as future work

1.10 Key Performance Indicators (Success Metrics)

Area	KPI	Target
Detection quality	AUC / F1 of Barrier 3 on held-out test set	AUC >= 0.95; F1 >= 0.90
Generalisation	AUC on unseen dataset/generator	Documented; target >= 0.80
False positives	Genuine images flagged as fake	<= 5% at the chosen operating point
Crypto integrity	Detection of tampering on registered images	100%
Watermark robustness	Bit accuracy after JPEG Q70 / 50% resize	>= 95%
Imperceptibility	PSNR / SSIM of watermarked vs original	>= 38 dB / >= 0.97
Performance	End-to-end verification time	< 10 s (95th percentile)
Usability	Task success in user test (n >= 10), SUS score	>= 90% success; SUS >= 75
Project delivery	Milestones met on time; test coverage of core modules	>= 80% on time; >= 70% coverage

1.11 Assumptions, Constraints and Proposed Technology Stack

- **Assumptions:** access to public datasets; availability of a GPU (local or cloud); the supervisor approves the scope.
- **Constraints:** 16-week academic schedule, student budget (open-source tools only), limited training compute.
- **Proposed stack (to be confirmed at design review):** Python, PyTorch, OpenCV; FastAPI backend; PostgreSQL; Redis (optional); Docker; Flutter or React Native mobile app; Android Keystore / iOS Secure Enclave; GitHub Actions for CI.

Part 2: Literature Review

2.1 Review Method

We searched for existing standards, research and products that address image authenticity, grouped into four families that match the four barriers: (1) provenance and cryptographic signing, (2) watermarking, (3) classical image forensics, and (4) learning-based manipulation and deepfake detection. For each family we describe how it works, what it does well, and where it falls short, then compare it with Image DNA. Sources are listed in section 2.7. Entries should be re-checked by the team for the latest versions and figures before final submission.

2.2 Provenance and Cryptographic Approaches

C2PA / Content Credentials. The Coalition for Content Provenance and Authenticity publishes an open specification for attaching a cryptographically signed manifest to media. The manifest records origin, edits, tools used and whether generative AI was involved, and each conforming edit adds a new manifest referencing the previous one, forming a provenance chain [1][2]. *Strengths:* open standard, industry adoption, tamper-evident, strong trust model based on X.509 certificates. *Weaknesses:* it proves provenance, it does not detect fakes; most images carry no credentials, and many pipelines and platforms strip metadata, so a manifest often does not survive to the viewer [3]. Absence of credentials is not evidence of forgery.

Secure-capture platforms (e.g., Truepic). Commercial products sign images at the moment of capture inside a controlled camera pipeline, so the image is trusted from creation. *Strengths:* strong trust at source. *Weaknesses:* only helps for images captured with the vendor's tooling; useless for the large existing pool of unsigned images. Image DNA's Barrier 4 follows the same idea in an open, student-scale form.

2.3 Watermarking

Classical frequency-domain watermarking (DWT, DCT, SVD) hides bits in transform coefficients and is fast, simple and fairly robust to compression [4]. **Neural watermarking** trains an encoder/decoder pair end-to-end with simulated distortions: HiDDeN [5] and StegaStamp [6] (robust even to printing and photographing) are representative. **Generative-model watermarks** such as Stable Signature [7] and Google's SynthID mark AI-generated images at creation. *Strengths:* invisible, works without a registry lookup. *Weaknesses:* only identifies images that were marked; can be attacked by regeneration, heavy cropping or screenshots; marking is cooperative, not universal.

2.4 Classical Image Forensics

Error Level Analysis (ELA) and tools such as FotoForensics highlight regions whose JPEG compression history differs from the rest of the image [8]; related techniques include noise-residual (PRNU) analysis, double-JPEG detection, copy-move detection and lighting inconsistency analysis. *Strengths:* no training needed, interpretable heat-maps. *Weaknesses:* need expert interpretation, easy to defeat by re-saving or resizing, high false-positive rates on images that were legitimately recompressed, and weak against modern AI-generated images.

2.5 Learning-Based Manipulation and Deepfake Detection

FaceForensics++ [9] introduced a large benchmark of face manipulations (Deepfakes, Face2Face, FaceSwap, NeuralTextures) and showed that CNN detectors such as XceptionNet [10] perform

strongly on seen manipulation types and compressed data. The **DFDC** challenge dataset [11] added scale and diversity. A recurring finding is **poor generalisation**: detectors trained on one generator degrade on others. Work such as **UnivFD** [12] (features from large pre-trained models) and **DIRE** [13] (reconstruction error from diffusion models) tackle generalisation to GANs and diffusion models. Commercial detectors (e.g., Sensity, Hive, Intel FakeCatcher for video) offer APIs or enterprise platforms, but they are closed, produce a probability only, and publish limited independent evidence of robustness. *Strengths:* can analyse a never-seen image. *Weaknesses:* probabilistic, dataset bias, an arms race with generators, vulnerable to adversarial perturbation and to compression/resizing.

2.6 Comparison Table

Solution / family	Approach	Strengths	Weaknesses	Relation to Image DNA
C2PA / Content Credentials [1][2]	Signed provenance manifest	Open standard, tamper-evident, edit history	No fake detection; metadata often stripped; needs tool adoption	Barrier 1 borrows signature concept; we add hash registry + AI fallback
Truepic-style secure capture	Sign at capture	High trust at source	Vendor-specific; only new, captured images	Barrier 4 secure capture (open, lightweight)
Classical watermarking [4]	DWT/DCT/SVD hidden bits	Simple, invisible, fast	Fragile to heavy edits/screenshots	Barrier 2 baseline
Neural watermarking [5][6][7]	Learned encoder/decoder	More robust, covers AI outputs	Cooperative; needs training; attackable	Optional Barrier 2 upgrade
ELA / FotoForensics [8]	Compression-artifact analysis	No training, visual	Expert needed; many false positives	Used as supporting feature/visualisation
FaceForensics++ detectors [9][10]	CNN on face manipulations	Strong on seen methods	Weak cross-generator generalisation	Core of Barrier 3 with improved evaluation
UnivFD / DIRE [12][13]	Pre-trained features / diffusion reconstruction	Better generalisation	Heavier compute; newer, less mature	Candidate backbones for Barrier 3
Commercial detectors	Closed AI APIs	Polished, scalable	Black box; cost; probability only	Benchmark / comparison only
Image DNA	**4 layers fused into one verdict**	**Complementary evidence; explainable; works with or without registry**	**Student-scale data/compute; AI layer remains probabilistic**	-

2.7 Gap Analysis and What Image DNA Adds

- Existing tools are **single-technique**; each fails exactly where another succeeds. Image DNA fuses four complementary signals so that one weak signal does not decide the outcome.
- Provenance tools say nothing about **unsigned** images and detectors say nothing definitive about **signed** ones. Image DNA handles both: cryptographic certainty when a registered original exists, statistical evidence when it does not.
- Most solutions return a bare score or a binary label. Image DNA returns an **explainable report** (per-barrier evidence, heat-maps, confidence).
- The focus on **identity documents, certificates and faces** in an accessible mobile experience for organisations such as universities and HR is under-served by research prototypes.

2.8 Lessons Applied to Our Design

- Evaluate **cross-dataset** (train on one source, test on another), because in-distribution scores overstate real-world performance [9][12].
- Use **pre-trained backbones** (EfficientNet/Xception/ViT) with transfer learning to cope with limited data and compute.
- Never treat missing provenance or watermark as proof of forgery [3]; map evidence to a graded verdict.
- Simulate real-world distortions (JPEG, resize, noise, screenshot) when training and testing both the watermark and the detector.
- Publish limitations openly: this builds trust and prevents misuse of the verdict.

2.9 Candidate Datasets

Dataset	Content	Use in project
FaceForensics++ [9]	Real and manipulated face videos (frames extracted)	Face-swap / reenactment training and testing
Celeb-DF	Celebrity real and deepfake videos (higher visual quality)	Face-manipulation training and cross-dataset evaluation
DFDC [11]	Large deepfake video dataset	Additional face-manipulation data, diversity
CASIA v2 / Columbia	Splicing and copy-move image forgeries	ELA + ResNet50 pixel-level tampering detection
FFHQ + generated sets (StyleGAN, diffusion)	Real faces and AI-generated counterparts	Synthetic image detection, cross-generator tests
Team-made document set	Own certificates / ID mock-ups, edited in Photoshop	Domain-specific testing; no real personal data

2.10 References

[1] C2PA, *Content Credentials: C2PA Technical Specification*, c2pa.org/specifications.

[2] Content Authenticity Initiative, *Understanding manifests* (Open-source SDK docs), opensource.contentauthenticity.org.

[3] Practitioner commentary on C2PA limitations (metadata stripping, absence of credentials), e.g. ppc.land/c2pa and guptadeepak.com glossary: C2PA (accessed 2026).

[4] I. J. Cox, M. L. Miller, J. A. Bloom et al., *Digital Watermarking and Steganography*, 2nd ed., Morgan Kaufmann, 2007.

[5] J. Zhu, R. Kaplan, J. Johnson, L. Fei-Fei, "HiDDeN: Hiding Data With Deep Networks," ECCV, 2018.

[6] M. Tancik, B. Mildenhall, R. Ng, "StegaStamp: Invisible Hyperlinks in Physical Photographs," CVPR, 2020.

[7] P. Fernandez et al., "The Stable Signature: Rooting Watermarks in Latent Diffusion Models," ICCV, 2023.

[8] N. Krawetz, "A Picture's Worth: Digital Image Analysis and Forensics," Black Hat, 2007 (Error Level Analysis); FotoForensics.com.

[9] A. Rossler et al., "FaceForensics++: Learning to Detect Manipulated Facial Images," ICCV, 2019.

[10] F. Chollet, "Xception: Deep Learning with Depthwise Separable Convolutions," CVPR, 2017.

[11] B. Dolhansky et al., "The DeepFake Detection Challenge (DFDC) Dataset," arXiv:2006.07397, 2020.

[12] U. Ojha, Y. Li, Y. J. Lee, "Towards Universal Fake Image Detectors that Generalize Across Generative Models," CVPR, 2023.

[13] Z. Wang et al., "DIRE for Diffusion-Generated Image Detection," ICCV, 2023.

Part 3: Requirements Gathering

3.1 Stakeholders and Their Needs

Stakeholder	Description	Main needs
Individual users	People checking a photo, certificate or ID received from someone else, or protecting their own images	Fast, simple check; clear verdict in plain language; privacy of uploads
Document issuers	Universities, training centres, companies issuing certificates	Register/sign issued documents; embed watermark; revoke; audit trail
Verifiers	HR departments, banks, notaries, recruiters, government offices	Reliable verification; explainable report to attach to a case file; API integration; bulk checks
Journalists / fact-checkers	Verify images before publication	Evidence per technique, heat-maps, confidence level
System administrator	Operates the platform	Monitoring, key management, user management, backups, model updates
Project team / supervisor	Builds and evaluates the system	Maintainable code, measurable results, documentation
Attackers (negative stakeholder)	Try to bypass or abuse the system	Considered in security requirements and threat model

3.2 Use Cases

ID	Use case	Actor	Summary
UC1	Register and sign an image/document	Issuer	Issuer uploads an original; system computes hash, signs it, embeds watermark, stores record; returns protected image
UC2	Verify an image	User / Verifier	User submits an image; system runs the four barriers and returns a verdict report
UC3	Capture a secure image	User (mobile)	App captures a photo and signs it on-device with a hardware-backed key
UC4	View and export report	User / Verifier	User opens per-barrier evidence and exports a PDF report
UC5	Revoke an issued image	Issuer / Admin	Marks a registered image as revoked; later verification shows revocation
UC6	Manage keys, users and models	Admin	Key rotation, role management, deploy a new model version
UC7	Verify via API	Organisation system	Third-party system calls REST API with an API key and receives JSON verdict

3.3 Functional Requirements

Priority uses MoSCoW: **M** = Must, **S** = Should, **C** = Could.

ID	Requirement	Priority
FR-01	The system shall accept JPEG and PNG images up to a configurable size (default 15 MB) from the mobile app, web client and API.	M
FR-02	The system shall validate file type and integrity and reject unsupported or corrupt files with a clear message.	M
Barrier 1: Cryptographic fingerprint		

ID	Requirement	Priority
FR-10	The system shall compute a SHA-256 hash of the image bytes.	M
FR-11	The system shall sign registered images with ECDSA (P-256) and store hash, signature, issuer, timestamp in the registry.	M
FR-12	The system shall verify a submitted image against the registry and report exact match / no match / revoked.	M
FR-13	The system shall compute a perceptual hash (pHash) to find near-duplicates of registered images (resized or recompressed).	S
FR-14	The system shall support revocation of registered images and key rotation.	S
Barrier 2: Invisible watermark		
FR-20	The system shall embed an invisible payload (issuer ID + record ID) into an image when it is registered.	M
FR-21	The system shall extract the watermark from a submitted image and report intact / partially damaged / absent.	M
FR-22	The watermark shall remain readable after JPEG recompression (Q >= 70) and moderate resizing.	M
FR-23	The system should localise regions where the watermark is missing (indicating local editing).	C
Barrier 3: AI & deepfake detection		
FR-30	The system shall classify an image as authentic vs manipulated/synthetic and return a probability score without needing the original.	M
FR-31	The system shall detect face swaps / face manipulation in images containing faces.	M
FR-32	The system shall detect fully AI-generated images (GAN/diffusion) as a separate score.	S
FR-33	The system shall generate a heat-map highlighting regions that influenced the decision.	S
FR-34	The system shall expose model version and evaluation metadata with each result.	S
Barrier 4: Mobile protection		
FR-40	The mobile app shall capture photos and sign them on-device using a hardware-backed key (Android Keystore / iOS Secure Enclave) where available.	M
FR-41	The app shall attest its integrity (Play Integrity / App Attest) and the backend shall check the attestation.	S
FR-42	The app shall detect rooted/jailbroken devices and warn or restrict sensitive actions.	S
FR-43	All communication shall use TLS 1.2+ with certificate pinning; sensitive data stored locally shall be encrypted.	M
Fusion, reporting and administration		
FR-50	The fusion engine shall combine barrier outputs into one verdict: Authentic (verified), Tampered, Suspicious, or Unverified, with a confidence score.	M
FR-51	The report shall list each barrier's result and explain in plain language why the verdict was reached.	M
FR-52	The user shall be able to export the report as PDF and view verification history.	S
FR-53	The system shall provide user authentication, role-based access (user, issuer, admin) and API keys.	M
FR-54	The admin shall be able to manage keys, users, and deploy new model versions.	S
FR-55	The system shall log every verification and administrative action in a tamper-evident audit log.	S
FR-56	The system shall support Arabic and English user interfaces.	S

Selected Deep Learning and Computer Vision Models

The AI part of the project (Barriers 2 and 3) is built on the following models and techniques:

- **Face Manipulation & Deepfake Detection (Barrier 3):** EfficientNet-B4 as the core backbone with transfer learning, preceded by MTCNN / RetinaFace for automated face extraction and alignment. The model is trained and evaluated on benchmark datasets (Celeb-DF / FaceForensics++).
- **Pixel-Level Tampering Detection:** Error Level Analysis (ELA) for compression-artifact inspection, combined with a ResNet50 classifier to detect local splicing, copy-move and Photoshop edits, using the CASIA dataset.
- **Signal-Based Watermarking (Barrier 2):** frequency-domain transformations (DWT-DCT-SVD) for robust, invisible payload embedding and extraction.

3.4 Verdict Definitions

Verdict	Condition (summary)	Meaning for the user
Authentic (verified)	Exact hash match with valid signature, not revoked	The file is byte-identical to what a trusted issuer registered
Tampered	Registered record found but hash/signature mismatch, or watermark shows local damage, or AI score very high	Strong evidence that the image was modified or generated
Suspicious	No registry match; AI score or forensic evidence above threshold but not conclusive	Treat with caution; seek the original or manual review
Unverified	No registry match, no watermark, AI score low	Nothing proves it authentic and nothing indicates forgery

3.5 Non-Functional (Quality) Requirements

ID	Category	Requirement	Measure
NFR-01	Performance	Verification of one image shall complete quickly under normal load	< 10 s at 95th percentile (images up to 5 MB); hash/signature check < 1 s
NFR-02	Accuracy	Detection quality shall meet the targets in Part 1	AUC >= 0.95 (in-distribution); FPR <= 5%; results reported with confidence intervals
NFR-03	Security	Private keys never leave secure storage; data encrypted in transit and at rest; protection against OWASP Top 10 and MASVS items	TLS 1.2+, AES-256 at rest, no critical findings in security review
NFR-04	Privacy	Data minimisation; uploaded images deleted after a configurable retention period unless the user chooses to store them; no use for training without consent	Retention policy enforced and tested
NFR-05	Usability	A first-time user shall verify an image without training; verdicts in plain language with colour and icon, not colour only	>= 90% task success; SUS >= 75
NFR-06	Reliability	Graceful degradation: if one barrier fails, others still run and the report states which one failed	Availability >= 99% during demo period; no crash on invalid input
NFR-07	Scalability	Stateless API and separate model service allow horizontal scaling	Supports 50 concurrent verifications in load test
NFR-08	Compatibility	Android 9+ and iOS 15+ (or current-2 versions); modern browsers; JPEG/PNG	Tested on at least 2 devices per platform
NFR-09	Maintainability	Modular code per barrier, documented API (OpenAPI), CI with automated tests	>= 70% unit-test coverage on core modules
NFR-10	Explainability	Every verdict shall be traceable to evidence from at least one barrier	100% of reports contain per-barrier evidence

ID	Category	Requirement	Measure
NFR-11	Accessibility and localisation	Support Arabic (RTL) and English; readable font sizes and contrast	Checked against WCAG 2.1 AA contrast
NFR-12	Auditability	All verifications and admin actions logged with timestamp and actor	Log entries immutable by normal users

3.6 Security Threats Considered

Threat	Example	Planned control
Image modified after signing	Change name on a certificate	Hash + signature (Barrier 1)
Watermark removal or screenshot laundering	Re-photograph the screen	Perceptual hash, AI detector, status "watermark absent" is not proof of authenticity
Fake / AI-generated face	Face swap on ID photo	Barrier 3 detection, face-region analysis
Forged capture	Upload a gallery image claiming it was captured live	Secure capture + app attestation (Barrier 4)
Stolen API keys / accounts	Abuse of verification API	Rate limiting, key rotation, RBAC
Key compromise	Private signing key leaked	Hardware/KMS storage, rotation, revocation list
Replay or man-in-the-middle	Intercept uploads	TLS with pinning, nonces and timestamps
Adversarial perturbation	Noise to fool the classifier	Basic robustness tests; documented limitation

3.7 Constraints and Assumptions

- Academic timeline of 16 weeks and open-source tooling only.
- Models are trained on public datasets; performance on real-world forgeries may be lower and will be reported honestly.
- Initial support for still images only; video is future work.
- Results are decision-support evidence, not legal proof.

3.8 Traceability: Objectives to Requirements

Objective	Functional requirements	Non-functional requirements
O1 Crypto fingerprint	FR-10 to FR-14	NFR-01, NFR-03
O2 Watermark	FR-20 to FR-23	NFR-01, NFR-02
O3 AI detection	FR-30 to FR-34	NFR-02, NFR-10
O4 Mobile protection	FR-40 to FR-43	NFR-03, NFR-08
O5 Fusion and report	FR-50 to FR-56	NFR-05, NFR-06, NFR-10, NFR-12

3.9 Acceptance Criteria (Examples)

- **AC-1:** Given a registered image, when one pixel is changed and the file is verified, then the verdict is Tampered and Barrier 1 shows a hash mismatch.
- **AC-2:** Given a watermarked image recompressed at JPEG Q70, when verified, then the watermark is reported intact or partially damaged and the payload bit-accuracy is at least 95%.

- **AC-3:** Given a face-swapped test image from the held-out set, when verified, then Barrier 3 returns a manipulation score above the decision threshold.
- **AC-4:** Given an unregistered genuine photo, when verified, then the verdict is Unverified (or Authentic-looking low score), never Authentic (verified).
- **AC-5:** Given a rooted device, when the user tries a secure capture, then the app warns and the report marks the capture as lower trust.